

	Success	Efficiency			Interaction quality					
	In-time	Action	Conversation	Overall	Factual grounding	Situational relevance	Actionable guidance	User intent uptake	Guidance conciseness	Overall
Assistant										
Frontier API models										
Gemini 3.8 Flash	30.6	32.7	73.7	55.1	<u>82.7</u>	79.6	68.1	76.2	89.0	<u>53.4</u>
GPT-6 Astra	<u>26.5</u>	33.0	<u>82.7</u>	<u>59.9</u>	86.5	89.4	78.5	97.1	<u>86.1</u>	63.5
Gemini 3.1 Pro	20.8	30.9	69.9	52.1	61.5	66.0	56.6	80.4	65.4	27.9
Real-time streaming API models										
GPT-Realtime 2.1	21.2	35.3	32.7	34.3	80.9	<u>87.7</u>	<u>72.6</u>	76.2	58.1	49.5
Gemini 3.8 Live	11.9	33.5	89.7	63.2	74.7	78.2	58.0	77.7	86.0	39.9
Open-weight non-streaming VLMs										
Muse Glimmer	23.8	42.5	0.8	19.6	59.2	75.3	56.8	65.0	41.3	27.3
Qwen3.8-27B	18.4	37.3	36.4	37.9	70.9	81.7	62.4	<u>85.9</u>	71.9	37.4
Kimi-VL-A3B-Thinking	15.2	30.7	24.0	27.1	34.2	39.4	29.1	39.0	26.0	5.7
InternVideo3-8B	14.5	38.0	59.7	50.2	65.7	45.3	35.0	41.7	68.9	20.9
Molmo2-8B	13.4	41.0	2.3	19.7	54.5	31.7	32.3	33.4	19.2	11.6
Open-weight streaming VLMs										
Qwen3-Omni-30B	26.2	<u>45.0</u>	0.0	20.7	65.8	63.8	51.2	54.3	67.6	25.4
MiniCPM-o 4.5	13.7	49.5	0.0	21.3	55.4	44.7	33.5	46.5	51.8	14.6
Proact-VL	11.3	43.1	71.3	58.1	66.4	25.6	17.2	28.4	44.5	15.8
StreamingVLM	10.9	39.4	77.1	59.5	75.8	25.0	34.8	34.4	22.9	24.5

Table 1 Visual assistance in CookSim. All scores are on a 0–100 scale (↑). Success and efficiency are three-run means; efficiency is measured on in-time successes, with persona-weighted overall scores. Rubrics use run 1 and the full 16-item overall score. Bold/underline indicate the best/second-best assistant in each column; dashes denote unavailable scores.